# Aperiodicity Certificates from Markers, Deflation, and Invariants: A One-File, Machine-Verifiable Proof Schema for 2D SFT Tilings with a Neurosymbolic Discovery Loop

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**Abstract**

We present a certificate-based framework for proving aperiodicity of two-dimensional tilings defined by finite local matching rules (subshifts of finite type, SFT). A certificate consists of (i) a finite set of *markers* establishing *recognizability* (unique macro-decomposition) at scale $k$; (ii) an *iterable deflation rule* forcing an infinite hierarchy; and (iii) optional *invariant obstructions* (e.g. height/cocycle constraints) enabling strengthened guarantees such as drift and limiting frequencies. We prove a generic aperiodicity theorem: infinite *phase-locking* induced by recognizability and iterability forbids any nonzero period. We further give a fully local, SAT-checkable validation protocol: each certificate clause is reduced to finite-window satisfiability queries, justified by compactness of SFTs. Finally, we propose a neurosymbolic search loop in which an ML proposer generates candidate certificates and a verifier accepts them or returns counterexamples for hard-negative mining.

## 1 Introduction

Aperiodic tilings are a canonical example of global structure enforced by local constraints. They connect logic and computability (domino problem undecidability), symbolic dynamics (2D SFTs), and geometric combinatorics. However, classical proofs of aperiodicity are often *bespoke*: each tileset needs tailored hierarchical arguments and custom invariants.

We propose to standardize a broad class of such proofs into finite *certificates* that are: (i) *verifiable* by a trusted checker (SAT/SMT/proof assistant), (ii) *discoverable* by guided search or learning (CEGIS-style), and (iii) *portable* across tilesets that share a hierarchical forcing mechanism.

**Contributions.**

- We formalize a certificate schema $\mathsf{Cert} = (M, \Phi, \mathcal{I})$ for 2D SFT tilings.
- We prove a *generic* theorem: recognizability + iterability + phase-locking $\Rightarrow$ no periodic tilings.
- We formalize the *finite-to-global* justification enabling SAT-checkable verification of certificate components.
- We describe a neurosymbolic discovery loop (ML proposer + verifier) for certificate synthesis.

# 2 Preliminaries

## 2.1 Wang tiles and SFTs

We work with Wang tiles (edge-colored unit squares) because they encode finite local constraints cleanly; the framework extends to polygonal tiles with matching rules by standard encodings.

**Definition 1** (Wang tiles)**.** Let $\mathcal{A}$ be a finite set of tile types. Each tile $a \in \mathcal{A}$ has colors $(N(a), E(a), S(a), W(a)) \in \mathcal{C}^4$ for a finite color set $\mathcal{C}$.

**Definition 2** (Valid tilings; 2D SFT)**.** A configuration is a map $x : \mathbb{Z}^2 \to \mathcal{A}$. It is *valid* if for all $(i, j) \in \mathbb{Z}^2$,

$$E(x(i, j)) = W(x(i + 1, j)), \qquad N(x(i, j)) = S(x(i, j + 1)).$$

Let $\mathcal{X} \subseteq \mathcal{A}^{\mathbb{Z}^2}$ denote the set of valid tilings. $\mathcal{X}$ is a (nearest-neighbor) 2D subshift of finite type (SFT).

## 2.2 Shift action and periodicity

**Definition 3** (Shift action)**.** For $v \in \mathbb{Z}^2$, define $(\sigma^v x)(p) := x(p + v)$.

**Definition 4** (Periodicity and aperiodicity)**.** A tiling $x$ is *periodic* if $\exists v \neq 0$ with $\sigma^v x = x$. The SFT $\mathcal{X}$ is *aperiodic* if $\mathcal{X} \neq \emptyset$ and contains no periodic tilings.

## 2.3 Local rules and forbidden patterns

Equivalently, an SFT can be defined by a finite set $\mathcal{F}$ of forbidden patterns on a finite shape (radius $r$). For Wang tiles, $\mathcal{F}$ is exactly the set of edge-mismatch adjacencies, hence radius 1.

**Definition 5** (Patch)**.** For a finite region $R \subset \mathbb{Z}^2$, a patch is a function $P : R \to \mathcal{A}$. A patch is *legal* if it appears as $x|_R$ for some $x \in \mathcal{X}$.

# 3 Certificate Schema

Fix an integer scale $k \geq 2$.

**Definition 6** (Aperiodicity certificate)**.** A certificate is $\mathsf{Cert} = (M, \Phi, \mathcal{I})$ where:

1. $M \subseteq \mathcal{A}^{[-r,r]^2}$ is a finite set of *marker patches* of radius $r$.
2. $\Phi$ is a *deflation rule* consisting of: (a) a method to infer a unique global *phase* $\alpha \in \{0, \ldots, k-1\}^2$ from marker occurrences, (b) a unique partition of each $x \in \mathcal{X}$ into $k \times k$ macro-blocks of phase $\alpha$, (c) a finite macro-alphabet $\mathcal{B}$ of macro-tile types (legal $k \times k$ blocks), (d) induced adjacency constraints forming a macro-SFT $\mathcal{Y} \subseteq \mathcal{B}^{\mathbb{Z}^2}$, and (e) an *iterability* claim: the construction repeats at all levels $k^n$.
3. $\mathcal{I}$ is optional: a locally checkable invariant obstruction (e.g. height/cocycle constraints) used to strengthen proofs or derive additional structure.

## 3.1 Marker predicate

Given $M$, define the local predicate

$$\mathsf{Marker}(x, p) \iff x|_{p+[-r,r]^2} \in M.$$

# 4 Recognizability from Markers

## 4.1 Macro-grid phase

**Definition 7** (Macro-grid phase)**.** A phase is $\alpha = (\alpha_x, \alpha_y) \in \{0, \ldots, k-1\}^2$. It partitions $\mathbb{Z}^2$ into blocks

$$B^{\alpha}_{i,j} := [ik + \alpha_x, (i+1)k + \alpha_x - 1] \times [jk + \alpha_y, (j+1)k + \alpha_y - 1].$$

We call the union of block boundaries the *macro-grid* of phase $\alpha$.

## 4.2 Soundness and completeness

Intuition: markers are like boundary "tags" that can only occur on macro-boundaries (soundness) and occur frequently enough to determine the phase uniquely (completeness).

**Definition 8** (Soundness (informal))**.** Markers are *sound* if in any valid tiling they can only occur at positions compatible with a single global phase $\alpha$ (e.g. lying on the macro-grid).

**Lemma 1** (Marker soundness)**.** *Assume there exists a phase $\alpha$ such that for every $x \in \mathcal{X}$ and every $p \in \mathbb{Z}^2$,*

$$\mathsf{Marker}(x, p) \implies p \text{ lies on the macro-grid of phase } \alpha.$$

*Equivalently, markers cannot occur strictly in the interior of any $B^{\alpha}_{i,j}$.*

**Lemma 2** (Marker completeness implies unique phase)**.** *Assume marker soundness (Lemma 1) and the following completeness condition: for every $x \in \mathcal{X}$ and every sufficiently large square window $W_L = [-L, L]^2$, the pattern of marker occurrences inside $W_L$ determines the same phase $\alpha$ (independent of the chosen window and location). Then every $x \in \mathcal{X}$ has a unique global macro-grid phase $\alpha$.*

*Proof.* Fix $x \in \mathcal{X}$. By completeness, choose $L$ large enough that marker occurrences in $W_L$ determine a phase; call it $\alpha_L$. Consider two overlapping large windows $W_L(p)$ and $W_L(q)$. Their intersection contains markers; soundness forces these markers to lie on a single macro-grid, hence the inferred phases must agree on the overlap. Therefore $\alpha_L$ is consistent across translations and is unique. Taking $L \to \infty$ yields a unique global phase $\alpha$. □

**Lemma 3** (Unique composition)**.** *Under Lemma 2, each $x \in \mathcal{X}$ admits a unique decomposition into $k \times k$ macro-blocks $B^{\alpha}_{i,j}$.*

*Proof.* Given the unique phase $\alpha$, the blocks $B^{\alpha}_{i,j}$ are uniquely defined by arithmetic. Any other decomposition into $k \times k$ blocks would correspond to a different phase $\alpha' \neq \alpha$, contradicting uniqueness of phase. □

# 5 Deflation and Iterability

## 5.1 Macro-alphabet and deflation map

**Definition 9** (Macro-alphabet $\mathcal{B}$)**.** Let $\mathcal{B}$ be the set of all legal $k \times k$ blocks that appear in some $x \in \mathcal{X}$ when partitioned by the unique phase $\alpha$. Each $b \in \mathcal{B}$ is identified with its full $k \times k$ interior assignment in $\mathcal{A}^{[0,k-1]^2}$.

**Definition 10** (Induced macro-SFT $\mathcal{Y}$)**.** Two macro-tiles $b, b' \in \mathcal{B}$ are horizontally compatible if their touching boundary columns agree in colors at the base level; similarly for vertical compatibility. Let $\mathcal{Y} \subseteq \mathcal{B}^{\mathbb{Z}^2}$ be the SFT enforcing these induced compatibilities (nearest-neighbor on macro-cells).

**Definition 11** (Deflation (factor) map)**.** Define $\Pi : \mathcal{X} \to \mathcal{Y}$ by mapping each macro-block to its macro-tile label:

$$\Pi(x)(i,j) := x|_{B^{\alpha}_{i,j}} \in \mathcal{B}.$$

**Lemma 4** (Deflation is well-defined and shift-commuting on $k\mathbb{Z}^2$)**.** *Under Lemma 3,* $\Pi$ *is well-defined and for all* $v \in k\mathbb{Z}^2$*,*

$$\Pi(\sigma^v x) = \sigma^{v/k}\Pi(x).$$

*Proof.* Shifting $x$ by $v \in k\mathbb{Z}^2$ shifts macro-block boundaries to macro-block boundaries without changing phase, hence the macro-tile at $(i,j)$ becomes the macro-tile at $(i + v_x/k,\, j + v_y/k)$. □

### 5.2 Iterability

A certificate must guarantee that the macro-structure repeats indefinitely across scales.

**Definition 12** (Iterable deflation)**.** The certificate $\Phi$ is *iterable* if for every $n \geq 1$ there exists a well-defined deflation map $\Pi_n : \mathcal{X} \to \mathcal{Y}_n$ producing macro-tiles of size $k^n \times k^n$ such that: (i) $\Pi_1 = \Pi$, (ii) $\Pi_{n+1}$ equals applying the $n$-level partition then grouping $k \times k$ macro-cells, and (iii) at each level the macro-grid phase is uniquely determined (phase-locking).

**Remark 1.** In practice, iterability can arise from strong self-similarity ($\mathcal{Y} \cong \mathcal{X}$ up to relabeling) or from a finite substitution system on a bounded set of macro-prototypes. Our theorem below uses only the existence of unique phases at all levels, not a particular substitution form.

## 6 Main Aperiodicity Theorem

**Theorem 1** (Aperiodicity from infinite phase-locking)**.** *Let* $\mathcal{X}$ *be a 2D SFT and fix* $k \geq 2$*. Suppose a certificate* $\mathsf{Cert} = (M, \Phi, \mathcal{I})$ *satisfies:*

1. *(**Recognizability**) Each* $x \in \mathcal{X}$ *has a unique decomposition into* $k \times k$ *macro-blocks (Lemma 3).*
2. *(**Iterability**) The deflation is iterable (Definition 12), giving unique phases at all levels* $k^n$*.*

*Then* $\mathcal{X}$ *contains no periodic tiling.*

*Proof.* Assume for contradiction that there exists $x \in \mathcal{X}$ and a nonzero period vector $v \in \mathbb{Z}^2 \setminus \{0\}$ such that $\sigma^v x = x$.

Consider level $n$ macro-tiles of size $k^n \times k^n$. By iterability, the level-$n$ macro-grid phase is uniquely determined for $x$. Since $x$ is invariant under shift by $v$, the entire set of level-$n$ macro-boundaries must also be invariant under shifting by $v$. But shifting by $v$ preserves a $k^n$-grid if and only if $v \in k^n\mathbb{Z}^2$ (otherwise block boundaries move to non-boundary lines).

Therefore, for every $n \geq 1$, we must have $v \in k^n\mathbb{Z}^2$. The only integer vector divisible by $k^n$ for all $n$ is $v = 0$, contradicting the assumption $v \neq 0$. Hence no periodic tiling exists in $\mathcal{X}$. □

## 7 Optional Invariants (Strengthening)

Theorem 1 already proves aperiodicity. Invariants add two benefits: (i) alternate proofs via obstruction (useful if phase-locking is subtle), (ii) extra structure (drift, frequencies), useful for ML and for dynamic properties of tiling spaces.

### 7.1 Height/cocycle invariant

**Definition 13** (Edge increments)**.** Assign integers to allowed edge matches:

- For each allowed horizontal match type $c$ (a specific color-pair match), set $h_E(c) \in \mathbb{Z}$.
- For each allowed vertical match type $c$, set $h_N(c) \in \mathbb{Z}$.

**Definition 14** (Path integral)**.** For a lattice path $\gamma$ from $p$ to $q$, define the signed sum $H_x(\gamma)$ of increments along edges traversed by $\gamma$.

**Lemma 5** (Curl-free implies potential)**.** *If for every legal $1 \times 1$ square in $\mathcal{X}$ the signed sum of increments around its boundary is zero, then $H_x(\gamma)$ depends only on endpoints. Consequently, there exists a potential $F_x : \mathbb{Z}^2 \to \mathbb{Z}$ such that $F_x(q) - F_x(p) = H_x(\gamma)$ for any path $\gamma : p \to q$.*

*Proof.* If the boundary sum is zero for every unit square, then any closed loop integral is zero (decompose the loop into unit squares). If $\gamma_1, \gamma_2$ are two paths from $p$ to $q$, then $\gamma_1 \cup \overline{\gamma_2}$ is a closed loop, so $H_x(\gamma_1) = H_x(\gamma_2)$. Define $F_x(p) := H_x(\gamma_{0\to p})$ for any reference path from 0 to $p$; well-definedness follows. □

**Lemma 6** (Periodicity forces a closing constraint)**.** *If $x$ is periodic with period $v$, then $\Delta(v) := F_x(p+v) - F_x(p)$ is independent of $p$ and additive on periods: $\Delta(v+w) = \Delta(v) + \Delta(w)$.*

*Proof.* Translation invariance of $x$ implies the multiset of edge matches in any translated path is identical, hence $H_x(\gamma) = H_x(\gamma + v)$. For any $p$, choose a path $\gamma : p \to p+v$; then $F_x(p+v) - F_x(p) = H_x(\gamma)$, which equals $H_x(\gamma + p')$ for any $p'$ by periodicity, hence independent of $p$. Additivity follows from concatenating a $v$-path and a $w$-path. □

**Proposition 1** (Drift obstruction under deflation (template))**.** *Suppose the certificate provides a height system such that, for macro-scale $k$ and some $\lambda \geq 2$,*

$$\Delta(kv) = \lambda\, \Delta(v) \quad \textit{for all } v \in \mathbb{Z}^2$$

*whenever $x$ admits the deflation structure and $F_x$ is consistent across scales. Then no nonzero period $v$ can exist unless $\Delta(v) = 0$. If additionally the certificate ensures $\Delta(v) \neq 0$ for all nonzero $v$ consistent with the macro-grid, then $\mathcal{X}$ is aperiodic.*

*Proof.* If $v$ is a period, then so is $kv$ at the macro level (period lifts across scale). By the relation $\Delta(kv) = \lambda\Delta(v)$, iterating yields $\Delta(k^n v) = \lambda^n \Delta(v)$. But $\Delta(k^n v)$ is a single integer determined by a bounded fundamental domain of the periodic tiling at scale $k^n$, forcing boundedness as $n \to \infty$ unless $\Delta(v) = 0$. Hence any nonzero period must satisfy $\Delta(v) = 0$. The final clause rules out all nonzero periods. □

**Remark 2.** Proposition 1 is an optional strengthening. In practice, the verifier checks the curl-free constraints locally and checks the drift/scaling identity on a finite set of macro-prototypes provided by $\Phi$.

## 8 Machine Verification via Finite SAT Checks

We now formalize why finite SAT checks can certify global properties in SFTs.

### 8.1 Compactness and finite witness

Let $\mathcal{A}^{\mathbb{Z}^2}$ carry the product topology with $\mathcal{A}$ discrete. Then $\mathcal{A}^{\mathbb{Z}^2}$ is compact by Tychonoff, and $\mathcal{X}$ is closed (defined by forbidden local patterns), hence compact.

**Lemma 7** (Finite witness principle for SFT extendability)**.** *Let $\mathcal{X}$ be an SFT defined by forbidden patterns of radius $r_0$. If there exists $x \in \mathcal{X}$ satisfying some global property $\mathcal{P}$ that is expressible as satisfaction of a set of local constraints on every finite window, then for every $L$ there exists a legal patch on $[-L, L]^2$ consistent with $\mathcal{P}$ restricted to that window. Conversely, if for some $L$ no patch on $[-L, L]^2$ can satisfy the local constraints for $\mathcal{P}$, then no global $x \in \mathcal{X}$ can satisfy $\mathcal{P}$.*

*Proof.* ($\Rightarrow$) Restriction of $x$ to any finite window yields a patch satisfying the restricted constraints. ($\Leftarrow$) Suppose patches exist for all $L$. Consider the inverse limit of consistent patches: by compactness (Kőnig's lemma / diagonal argument) one can extract a convergent subsequence of patches whose limit is a global configuration in $\mathcal{A}^{\mathbb{Z}^2}$. Since the forbidden constraints are local and satisfied on every finite window, the limit lies in $\mathcal{X}$. Likewise, the local constraints defining $\mathcal{P}$ are satisfied on all windows, hence $\mathcal{P}$ holds globally. $\square$

### 8.2 SAT encoding

Given a finite window $W = [0, L-1]^2$, introduce Boolean variables $x_{i,j,a}$ for $(i, j) \in W$ and $a \in \mathcal{A}$, meaning tile $a$ is placed at cell $(i, j)$. Constraints:

- Exactly one tile per cell: $\sum_{a \in \mathcal{A}} x_{i,j,a} = 1$.
- Color matches: forbid assignments where neighboring tiles mismatch.
- Marker placement constraints: enforce a specific patch equals one of the marker patterns at a location, or forbid it.
- Competing phase constraints: encode existence of two distinct global phases consistent with marker rules within the window.

A SAT solver returns either satisfiable with a witness patch or UNSAT.

### 8.3 Verifier checks

Given $(\mathcal{A}, \text{rules})$ and a proposed certificate $(M, \Phi, \mathcal{I})$:

**(V1) Marker soundness.** For each marker $m \in M$ and each illegal offset modulo $k$, query SAT: does there exist a legal patch on a window $W$ in which $m$ occurs in an interior location inconsistent with phase $\alpha$? Require UNSAT for all such queries. By Lemma 7, UNSAT implies global impossibility.

**(V2) Unique phase / unique composition.** Encode two distinct phases $\alpha \neq \alpha'$ (finite domain variables) and enforce that marker occurrences in $W$ are consistent with both. Require UNSAT to prove no patch supports two phases; apply Lemma 7 to conclude global uniqueness.

**(V3) Deflation consistency.** Check each macro-prototype is a legal $k \times k$ block and that macro-adjacency implies base-level boundary matches. These are finite checks over a finite set of prototypes.

**(V4) Invariant checks (optional).** Curl-free constraints are local checks over legal $1 \times 1$ squares; drift identities are checks over macro-prototypes.

# 9 Neurosymbolic Certificate Discovery

We cast certificate discovery as a counterexample-guided loop: an ML proposer suggests $(M, \Phi)$, and the verifier either accepts or returns counterexamples.

## 9.1 Why ML helps

The certificate space is combinatorial: choosing markers is selecting rare but decisive local patterns; choosing deflation is proposing a consistent segmentation and prototype clustering. This resembles program synthesis and structure discovery, where learning can propose high-probability candidates and solvers provide correctness.

## 9.2 CEGIS-style loop

**Algorithm 1** ML Proposer + SAT Verifier (CEGIS-style)

**Require:** Rules defining $\mathcal{X}$, scale $k$, model parameters $\theta$

```
for t = 1, 2, ... do
    Sample legal patches {P_i} from X (SAT sampling / MCMC / inflation)
    (M_t, Φ_t) ← ProposeCert_θ({P_i})
    RESULT ← Verify(M_t, Φ_t)
    if RESULT = PASS then
        Store (M_t, Φ_t); optionally search for invariant I
    else
        Extract counterexample patch Q
        Add Q to hard negatives; update θ to reduce recurrence
    end if
end for
```

# 10 Experimental Plan (Workshop-ready)

**Benchmarks.** (i) Known aperiodic Wang tilesets and hierarchical constructions, (ii) parametric perturbations (recolor / swap constraint subsets), (iii) random instances filtered for non-emptiness.

**Metrics.** PASS rate, time-to-certificate, SAT calls, certificate size ($|M|$, $|\mathcal{B}|$), and cross-family generalization.

**Ablations.** Random search vs ML proposer; markers-only vs markers+deflation vs full (+ invariant); varying $k$.

# 11 Limitations, Safety, and Ethics

**Limitations.** SAT checks may scale poorly as window size grows. Some tilings require invariants beyond the height template. Our theorem gives a sufficient condition (infinite phase-locking), not a necessary characterization.

**Safety and ethics.** The main impact is positive: verifiable reasoning artifacts and reproducible proof pipelines. Compute cost and energy usage should be reported; the method does not inherently enable harmful capabilities beyond general-purpose SAT/ML.

## 12 Conclusion

We proposed a one-file, certificate-based proof schema for aperiodicity in 2D SFT tilings: markers give recognizability, deflation yields infinite hierarchy, and phase-locking forbids periodicity. A finite SAT-based verifier checks certificates with compactness guarantees, and an ML proposer can search certificate space with counterexample-guided refinement. This bridges classical tiling theory with verifiable neurosymbolic proof synthesis.

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